

Developing a Novel Multi-fusion Brain-Computer Interface (BCI) System with Particle Swarm Optimization for Motor Imagery Task

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Abstract—In this paper, we develop a novel multi-fusion brain-computer interface (BCI) based on linear discriminant analysis (LDA) to deal with motor imagery (MI) classification problem. We combine filter bank and sub-band common spatial pattern (SBCSP) to extract features from EEG data in the preprocessing phase, and then LDA classifiers are applied to classify brain activities to identify either left or right hand imagery. To further foster the performance of the proposed system, a fuzzy integral (FI) approach is employed to fuse information sources, and particle swarm optimization (PSO) algorithm is exploited to globally update parameters in the fusion structure. Consequently, our experimental results indicate that the proposed system provides superior performance compared to other approaches.

Keywords—Brain-Computer Interface (BCI); Motor Imagery (MI); Sub-Band Common Spatial Pattern (SBCSP); Fuzzy Integral (FI); Particle Swarm Optimization (PSO)

I. INTRODUCTION

A brain-computer interface (BCI) is a communication system that has the ability to describe a user's intent even without any peripheral muscular activity involved [1]. BCI systems enable a subject to send commands only by means of brain activity. Such interface has huge potential to become solution for people affected by a number of motor disabilities. Among all alternatives, Electroencephalography (EEG) has been widely adopted as the brain activity identifier to build BCI systems. EEG signals are generally categorized into five specific frequency bands, namely delta: <4Hz, theta: 4-7Hz, alpha: 8-12Hz, beta: 12-30Hz and gamma: >30Hz [2]. Previous studies have found that imagination of movement alters the sensorimotor rhythms (SMR) concerning oscillations in brain activity in somatosensory and motor areas. In particular, alpha band activity in sensorimotor area and beta band activity are important indicators for motor imagery. Alpha band activity in the sensorimotor area is also called mu activity.

The essence of a BCI system includes real-time signal measurement and process of signals to detect certain patterns in order to send precise commands. In order to achieve this, signal processing involves three stages: preprocessing, feature extraction and classification. Preprocessing aims at simplifying the following processing operations by arranging the input signals in a uniform format, removing any noise or outlier and

preserving all relevant information. Feature extraction can be seen as an advanced step in preprocessing. In this step, only the most important features are extracted and preserved in order to improve the system in the following two perspectives: 1) enhance computational efficiency, and 2) enhance classification performance. In the final stage, classification algorithms are applied to detect different patterns in order to transform brain signals into commands.

The following sections are arranged as follows. In section II, the motor imagery experiment and the proposed system are introduced. In section III, results are presented and some issues are discussed. Finally in section IV, a brief conclusion is made and future works are suggested.

II. MATERIALS AND METHODS

Our proposed algorithm is divided into four stages; the first three correspond to the previously mentioned three stages, including preprocessing, feature extraction, and classification, of a conventional BCI system, and the additional stage in our proposed system utilizes fusion approach as depicted in Fig. 1. In order to measure the brain dynamics and enhance classification accuracy and stability, we divide EEG signals into five frequency bands and apply an ensemble of LDAs to classify each of the frequency bands. An Information fusion scheme is applied to fuse classifier results. Particle swarm optimization (PSO) is utilized to tune classifier weights during fusion stage.

A. Filterbank and sub-band common spatial pattern (SBCSP)

Common spatial pattern (CSP) is a feature extraction method used in this study, and its effect on overlapping data is shown in Fig. 2. Although it is effective to manually find the best band for each subject, it is also time consuming and the result is hardly repeated by other research groups. Here, we use a filter bank to decompose the EEG signals into 4 sub-bands, which include delta, theta, alpha, and beta bands. Additionally, a cross-band frequency feature is preserved in order to capture the interaction among different bands. This method refers to sub-band CSP which has been proven to achieve similar result as finding proper bands manually for each subject. That is, SBCSP achieves higher accuracy than CSP and provides much more efficiency and consistent performance than finding proper bands for subjects manually [3].

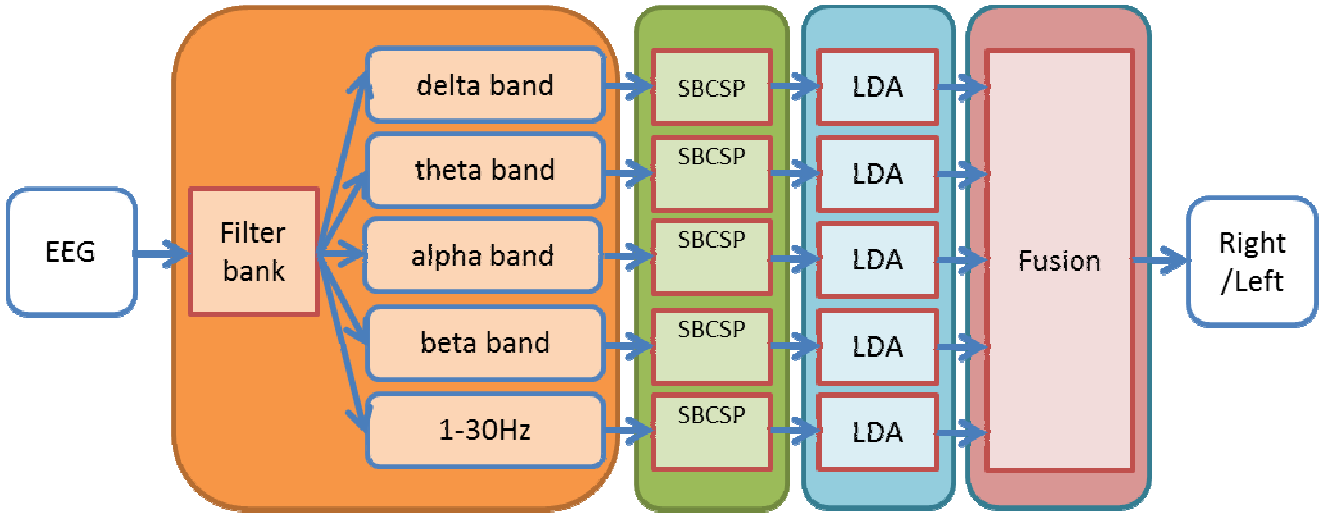


Fig.1 Proposed system architecture.

B. Linear discriminant analysis (LDA)

LDA [4] is a well-known classification method which projects data in a new space using a projection: $y = w^T x$, which minimizes the within class scatter and maximizes the scatter between classes. There are two classes as shown in Fig. 3. In this black bar, when the data points gain higher score over threshold defined by user, then we can regard that point as class 1. On the other hand, if the score of data points is less than our defined threshold, then it will belong to class 2.

C. Fusion

In this sub-section, three different types of fusion schemes, including weighted summation, Sugeno integral, and Choquet integral are introduced below. When the summation is applied, we simply sum up the posterior probabilities from each LDA's for right movement and left movement, and assign the instance to the class with the higher summation value.

The Sugeno integral [5, 6] is a type of integral with respect to a fuzzy measure. The Sugeno integral over a crisp set A of a function h with respect to a fuzzy measure g is defined by

$$S_g(h) = \int_A h(x) \circ g = \sup_{\alpha \in [0,1]} [\min(\alpha, g(A \cap F_\alpha))] \quad (1)$$

where $F_\alpha = \{x | h(x) \geq \alpha\}$.

When applying to a fusion problem, the function h is

generally referred to the classifier output scores and the fuzzy measure g is the confidence or the weight we give to each classifier.

On the other hand, the Choquet integral [7], created by the French mathematician Gustave Choquet, is another type of integral with respect to a fuzzy measure. It is defined as

$$C_g(h) = \sum_{i=1}^n [h(x_i) - h(x_{i-1})]g(A_i) \quad (2)$$

where $A_i = \{x_i, \dots, x_n\}$.

As aforementioned, when applying to a fusion problem, h is referred to classifier outputs and g is the weight for each classifier.

D. Particle swarm optimization (PSO)

The PSO algorithm [8] is one of well-known swarm intelligence technology. It was developed by imitating movement of organisms in a bird flock or fish school. The purpose of PSO is to optimize a model by iteratively trying to improve a candidate solution with regard to a given measure of quality. The PSO algorithm involves two critical steps:

1. Initialize a population of particles with random distribution within the desired range of the search space.
2. Update the particle position and velocity:

$$v_{id} \leftarrow \omega v_{id} + \phi_p r_p (p_{id} - x_{id}) + \phi_g r_g (g_d - x_{id}),$$

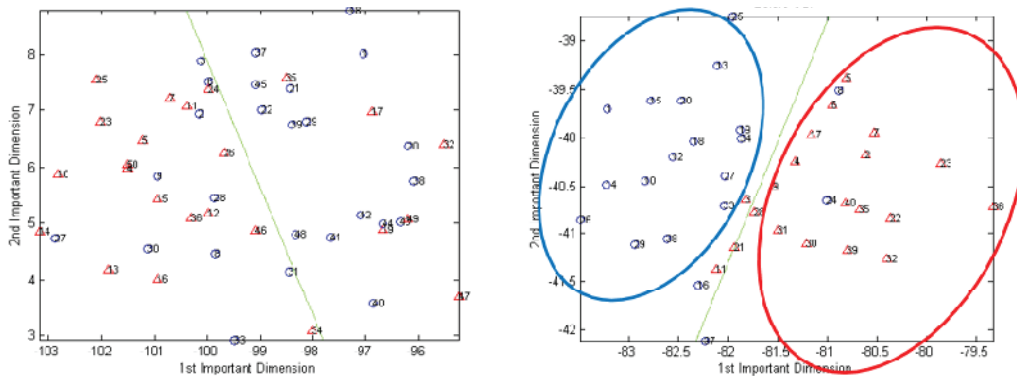


Fig.2 Scatter map before (left) and after (right) CSP projection.

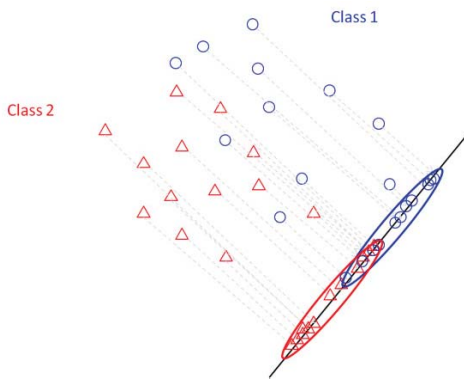


Fig.3 The LDA algorithm projects data to the hyper-plane that maximizes the between class scatter and minimizes the within class scatter.

$$x_i \leftarrow x_i + v_i \quad (3)$$

This procedure provides us to find optimal weights for each information source to obtain global optimized output during the training phase.

E. Experimental design

In this study, we recruited four subjects to collect the EEG signals in the MI task. During the procedure, the participants were trained to imagine right hand and left hand movement, respectively. All EEG data were recorded using the Mindo 4S

TABLE I. CLASSIFICATION RESULT FOR BASE CLASSIFIERS AND DIFFERENCE FUSION TECHNIQUES.

	Area Under Curve of ROC	
Component LDA's	LDA_delta	0.897±0.082
	LDA_theta	0.888±0.083
	LDA_alpha	0.876±0.086
	LDA_beta	0.876±0.083
	LDA_all band	0.891±0.082
Fusion Techniques	Summation	0.963±0.046
	Sugeno	0.947±0.055
	Choquet	0.963±0.045

[9] proposed in our previous study, a portable EEG recording device, with a 4-channel spring-loaded dry sensor. The sampling rate is at 512Hz and later down-sampled to 100Hz.

Next, in the proposed system, weighted summation, Sugeno integral, and Choquet integral are applied as fusion techniques to fuse the five base classifiers, delta, theta, alpha, beta and all-band LDA's. The weight of each classifier in Sugeno and Choquet integral is equally set to 1/5. Afterwards, the PSO algorithm would be applied to update these weights.

III. RESULTS AND DISCUSSIONS

All performance is evaluated in terms of area under receiver operating characteristic curve (AUC of ROC). As shown in Table I, each fusion technique outperforms single classifier, which the proposed fusion architecture exhibits not only a higher AUC value but also a lower standard deviation. In comparison with the three fusion techniques, summation and Choquet integral outperform Sugeno integral. Furthermore, we

applied PSO to update the weights for classifiers in Sugeno and Choquet integral. As shown in Table II, after tuning the weights, both the integrals obtain a significant improvement. Not only the AUC is improved but also the standard deviation is reduced, suggesting that the system is performing at higher accuracy and better stability.

IV. CONCLUSIONS AND FUTURE WORK

TABLE II. CLASSIFICATION RESULT FOR SUGENO AND CHOQUET INTEGRAL AFTER PSO TRAINING.

Fusion Techniques	Area Under Curve of ROC	Before PSO	After PSO
Sugeno		0.947±0.055	0.981±0.028
Choquet		0.963±0.045	0.985±0.024

In this study, we develop an innovative and reliable BCI system, in which applying SBCSP, LDA and fusion mechanism, for MI recognition task. Furthermore, after adopting Choquet integral along with PSO algorithm as a training algorithm at the fusion level, we obtain the highest AUC. By means of use of a portable EEG collecting device, Mindo 4S, and the proposed algorithm, we can develop practical BCIs in real-world scenarios. In the future, we aim to develop a more advanced BCI system that is capable of classifying multiple-class problems and enables our system to execute more complex cognitive tasks.

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